State of the art and objectives

Controlling light-matter interactions via carefully optimized materials is the key to advancement within a number of contemporary science disciplines and technologies including lasers, photovoltaics, photocatalysis, plasmonics, optical metamaterials, and quantum light sources. These different fields are all facing a common challenge: the photo-physical properties of the employed materials are determined by a complex interplay between photo-excited carriers, localized defect states, material interfaces, and lattice vibrations (phonons) which makes it difficult to predict and rationalize their interaction with light. For this reason, traditional development of photoactive materials is a slow process where well-known materials are optimized based on empirical considerations.

The development of paradigm-shifting technologies based on novel materials with new functionalities must involve radically new approaches to material optimization that are based on systematic and rational design. For this to happen, a crucial first step is to establish a stronger link between the microscopic structure of a material (at the atomic and electronic levels) and its observable photo-physical properties. LIMA will establish this link by developing ground breaking quantum mechanical methods to quantify and predict light-matter interactions in real materials under operating conditions from first principles. The methods will be used to address some of the most outstanding challenges within the fields of photovoltaics (PV), plasmonics, photonic metamaterials, and single-photon emitters (SPE), and ultimately lead to the development of novel and untraditional materials with rationally designed light-matter interactions.

Although the function and purpose of the photoactive materials is different for each application, there are a number of common factors that are critical to their performance. Of particular importance are the detrimental non-radiative processes mediated by defects and phonons that lead to efficiency losses in PV cells[1], reduce the quantum efficiency of SPEs[2], and cause Ohmic losses in the metallic components of plasmonic devices[3]. Other common critical factors are the size and nature of the electronic band gap, the line-up of band edges at interfaces, and the position of electronic defect levels inside the band gap. *Mitigating non-radiative losses and engineering the electronic states via rational computer aided materials design to unlock new opportunities for controlling light-matter interactions, constitutes the overarching idea of LIMA.*

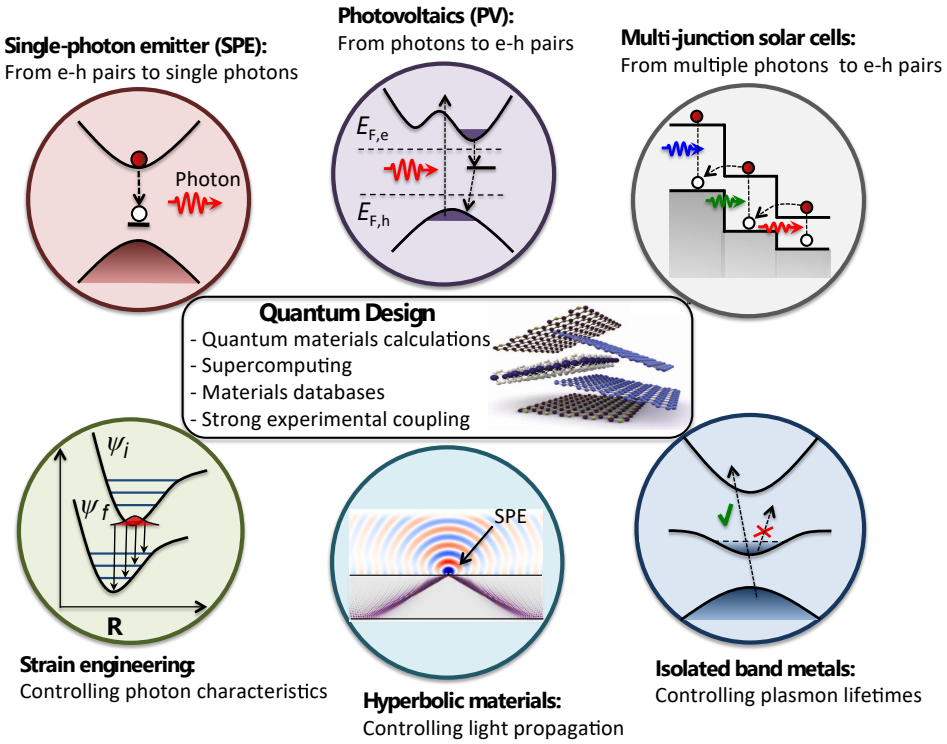


Figure 1. LIMA will address several aspects of light-matter interactions: From non-radiative loss processes in metals and semiconductors to photonic materials with peculiar light dispersion to single-photon emission. Combining powerful quantum materials simulations with systematic use of materials databases and strong experimental collaborations, the unique properties of 2D materials and van der Waals heterostructures will lead to novel materials for low-loss plasmonics, multi-junction solar cells, hyperbolic materials, and single-photon emitters.

In terms of materials, *LIMA* will focus mainly (but not exclusively) on the rapidly expanding class of atomically thin two-dimensional (2D) materials[4,5] which exhibit unique opto-electronic properties. The possibility of combining different 2D materials into van der Waals heterostructures (vdWHs)[6] provides a unique platform for controlling light-matter interactions with atomic scale precision, and offers an enormous amount of possibilities to broaden the versatility of 2D materials and to achieve superior and unusual material properties. But the enormous flexibility also posses a challenge: A vdWH containing 20 atomic layers picked from a pool of ten different 2D materials results in 10^{20} different stacking combinations. In reality there are more than 1000 different 2D materials available[7] and the number of possible combinations essentially endless. Obviously, rational and efficient methods to guide the design are required to fulfill the potential of vdWHs.

In the following I describe the State of the art within LIMA's focus areas (A-E). The research objectives are stated after each section and are based on the identified gaps/limitations of the State of the art.

A. First-principles calculations based on density functional theory (DFT) or many-body perturbation theory have been pivotal for our understanding of the chemical, thermodynamic, and electronic properties of materials close to their ground state. However, first-principles calculations of non-radiative recombination processes and photo-carrier lifetimes in real materials remains a largely unexplored field. Only recently, a few first principles calculations of non-radiative processes have been reported including carrier capture rates due to a substitutional defect in GaP and GaN[8], direct and phonon-assisted Auger scattering rates in GaN[9], phonon-mediated carrier relaxation rates[10] and plasmon lifetimes in noble metals[11]. These studies showed that it is feasible to obtain a detailed microscopic insight into non-equilibrium photophysical processes via the calculation of non-radiative relaxation rates from first principles.

Objective: *I will develop and implement a general framework for calculating non-radiative relaxation processes from first principles. Three different recombination processes will be considered, namely Shockley-Read-Hall (SRH) carrier trapping, defect-assisted and phonon-assisted Auger scattering. Additionally impact ionization will be considered as it represents a way to overcome the Shockley-Queisser efficiency limit of solar cells* ***(WP1)****.*

B. Photovoltaics (PV) is likely to become one of the most important sources of renewable energy in the future. Current PV technology depends on absorbers that have low efficiency (amorphous silicon), high cost (crystalline Si, III-V semiconductors), limited abundancy (In, Ga in CIGS or Te in CdTe) or could be toxic (Cd in CdTe)[1]. Organic-inorganic halide perovskites are promising candidates but they require toxic lead (Pb) to reach meaningful efficiencies and suffer from serious stability issues[12]. New materials are urgently needed to meet the upcoming terawatt challenge and to enable more flexible cell designs targeting different applications and operating conditions. Computational discovery of new PV materials has so far mainly focused on optimizing the properties of the pristine crystals (stability, band gaps, light absorption, etc.)[13,14]. While these are necessary conditions they are not sufficient because PV performance for all but the most crystalline materials (c-Si and GaAs) is determined by more complex factors like defect formation[15] and non-radiative recombination. The key to advancing PV technologies (as well as light emitting diodes and lasers) *via* computational materials discovery is to establish a stronger link between the microscopic structure of a material (at the atomic and electronic levels) and the efficiency limiting loss processes. For using nanostructured materials to control light-matter interactions it is further important to understand how reduced dimensionality influences the loss processes. Recent optical pump-probe experiments have revealed very short carrier lifetimes in 2D semiconductors[16], presumably due to efficient defect assisted non-radiative recombination, but a microscopic understanding is still missing.

Objective: *I will apply the methods developed in WP1 to calculate carrier lifetimes and open circuit voltages (V_{oc}) for a set of experimentally well-characterized PV materials taking phonons and realistic defect concentrations into account. On this basis, simple descriptors for "defect tolerance" and PV performance will be identified. Similar studies will be conducted for a set of 2D materials to investigate how the geometric/quantum confinement and reduced screening manifest in the carrier lifetimes and recombination dynamics.* ***(WP2).***

C. Multi-junction solar cells represent a particularly interesting class of PV devices with a great but largely untapped potential. They offer a high degree of flexibility in terms of cell design and could boost the efficiency dramatically from the theoretical limit of 33% for a single cell to 70% for a fully optimized multi-junction cell[17]. Of crucial importance for the efficiency of a multi-junction cell is that the band gaps and band alignment of the different absorber layers be carefully matched. The band alignment at the buried interfaces

is extremely difficult to control or even measure experimentally. Consequently, current multi-junction cells (efficiencies beyond 40% are made using highly advanced epitaxial growth of well characterized, but expensive, III-V semiconductor materials[18]. To make multi-junction cells competitive, there is a need for a completely different design based on cheaper materials that can be combined more easily. To this end vdWHs are promising candidates. Proof-of-concept experiments have recently demonstrated vdWH-based light emitting diodes[19], ultrafast photodetectors[20], and high mobility field effect transistors[21], but device performances are still far from the technological state of the art. Due to the weak interlayer forces the interfaces between different 2D materials, can be atomically sharp along the stacking direction and, in contrast to conventional covalently bonded interfaces, their formation does not introduce additional defects. There is in principle no restriction on the type of 2D materials that can be combined, making vdWHs ideal for band structure engineered devices such as multi-junction solar cells. The growing set of available 2D materials and endless possibilities for combining them calls for computational methods that can predict the electronic properties of realistic vdWHs for rational design of their interactions with light. Such methods do not exist today. It has been established that many-body effects are particularly important in 2D materials due to the weak dielectric screening, and the GW self-energy method has been demonstrated to yield excellent band structures of freestanding[22] and adsorbed monolayers[23]. However, due to their large size and complex structure, many-body band structure calculations for general vdWHs have so far been ruled out.

Objective: *Building on my previous work on dielectric screening and self-energy effects in vdWHs[24,25], I will develop a powerful multi-scale method based on the many-body GW method for accurate band structure calculations of general vdWHs containing thousands of atoms in a unit cell. The method will be used to identify specific stacking combinations for optimal light harvesting and minimal thermalization losses. I will also establish the fundamental limitations for atomic-scale band structure engineering with 2D materials.* ***(WP3).***

D. Plasmonics utilises the sub-wavelength confinement of light on metallic surfaces in the form of plasmon-polaritons to enable a range of applications including negative index materials[26], energy conversion[27], and imaging[28]. However, in practice many of these applications are challenged by optical losses occurring in the metallic components[29,30]. Previous searches for plasmonic materials in the optical regime have explored the alkali-noble intermetallics[31] and transition metal nitrides[32], but so far no viable alternatives to the noble metals have been found, and the detrimental losses have appeared to be an unavoidable property of any conducting material. The main decay mechanism of plasmons is Landau damping, i.e. creation of electron-hole pairs of either intra- or interband type. For plasmons in the technologically interesting near-infrared to visible regime, the annihilation of a plasmon must involve the creation of an electron-hole pair with the same energy as the plasmon. Thus the ideal plasmonic material is an isolated band metal (IBM), i.e. a material that has one, or several, partially occupied bands that are isolated from all higher and lower lying bands by finite energy gaps[33]. If the energy gaps are large enough and the plasmon energy falls in the gap between intra- and interband transitions where no electron-hole pairs exist, the non-radiative decay of the plasmon would be strongly suppressed. To my knowledge, no IBMs are currently known. This could indicate that IBMs do not exist. However, I have recently discovered that some of the known van der Waals layered materials that can be exfoliated to monolayers or thin films, have isolated, or nearly isolated, metallic bands. These materials by themselves have potential to significantly outperform the best presently known plasmonic metals. Additionally, combined with another recent discovery from my group, namely that such layered metals can exhibit the unusual property of hyperbolic light dispersion, the layered IBMs open unique possibilities for controlling light-matter interactions at the atomic level.

Objective: *I will perform an in-depth study of the optical properties of my already identified IBMs in their layered bulk and monolayer forms, and evaluate their potential for plasmonics via their figures of merit (plasmon lifetimes and propagation length, spatial confinement, etc.). Further IBMs will be discovered by screening materials databases and by subsequent combinatorial chemical substitution applied to the materials resulting from the screening.* ***(WP4)***

E. Single-photon emitters (SPE) play a central role in many leading quantum technologies including quantum communication and quantum information processing[34,35]. The ideal SPE delivers individual photons at a high rate (‘on demand’) with a narrow frequency distribution at room temperature and is stable over long time periods. Despite years of research on a variety of different material systems, the ideal SPE is yet to be discovered[2]. Color center-based SPEs, produce photons by spontaneous decay of discrete electronic excitations localized around point defects (native or impurities) in a host crystal. While the use of a solid state host has many advantages, it also presents significant challenges. The complex environment causes fluctuations in the frequency and phase of the emitted photons leading to spectral broadening and reduced indistinguishability. In addition, the collection of photons emitted from the interior of a bulk crystal can be

difficult. The most widely used solid state SPE is the Nitrogen-Vacancy (NV) center in diamond. The NV center is highly stable, which is an important reason for its popularity, but it suffers from a finite dipole moment making it sensitive to stray fields from the environment and a strong coupling to local lattice vibrations, which broadens the room temperature emission spectrum resulting in a reduced quantum efficiency of photons emitted in the sharp zero phonon band[36]. As quantum technologies continue to mature, the development of better SPEs becomes more and more urgent. Recently, a number of 2D materials have been shown to host SPEs[37,38,39]. The origin of the SPE was ascribed to excitons bound to defects or transitions between discrete defect states deep within the band gap; however, the crystal- and electronic structure of the 2D quantum emitters are not known and many open questions concerning their nature and function remain. Compared to bulk systems, defects embedded in a 2D material are much easier to create, control, and characterize. Moreover, integration of 2D SPEs with cavities, photonic waveguides, or plasmonic materials, is also much more straightforward. Thus 2D materials offer a fascinating platform for quantum photonics waiting to be explored.

Objective: *I will perform a computational screening of a large number of color centers in different 2D semiconductors/insulators and evaluate their potential as SPEs based on a series of carefully chosen design criteria, with the aim of identifying optimal SPEs targeting different quantum technologies* ***(WP5).*** *Focusing on the most stable color centers identified, I will explore the possibility of tuning the SPE characteristics by strain engineering or by modifying the photonic modes around the SPE by means of natural hyperbolic 2D materials.* ***(WP6)*** *A realization of a novel electrically controlled on-demand 2D SPE with hyperbolic electrodes (electro-hyperbolic SPE), based on the computationally designed materials, will be pursued in collaboration with my quantum optics and 2D nanoelectronics experimental collaborators.* ***(WP7)***

Why me? I have extensive experience with first-principles method development and computational materials design. During the past three years I have made substantial contributions to the description and understanding of excitons, plasmons and quasiparticles in 2D materials and have published close to 30 papers on the topic (PRL, Nano Letters, PRB, JPC, and a comprehensive review article to appear in *2D Materials*). I am developing a computational database with structural and electronic properties of 2D materials (both experimentally known and hypothetical structures not yet synthesized), see Fig. 2. The database has made a clear impact on the community and the paper has received 40 citations 1.5 year after publication[40].

My group is co-developing and maintaining the GPAW open source electronic structure code[41,42] and the Computational Materials Repository (CMR)[43] for storing and managing computed materials properties. I am part of a Danish Centre of Excellence on Nanostructured Graphene (CNG)[44], which has equipped me with a strong local and international network in 2D materials research. I am connected to the EU Graphene Flagship via the CNG. I am also a PI of the Centre of Excellence for Novel Materials Discovery (NOMAD)[45], which develops database infrastructure and Big Data analytics tools for materials science. Finally, I participate in the Center for the Science of Sustainable Fuels and Chemicals (V-SUSTAIN), which among other activities develops photo-absorbers and photo-catalysts for water splitting. The funding I receive from these centers corresponds roughly to 1 person/year/center.

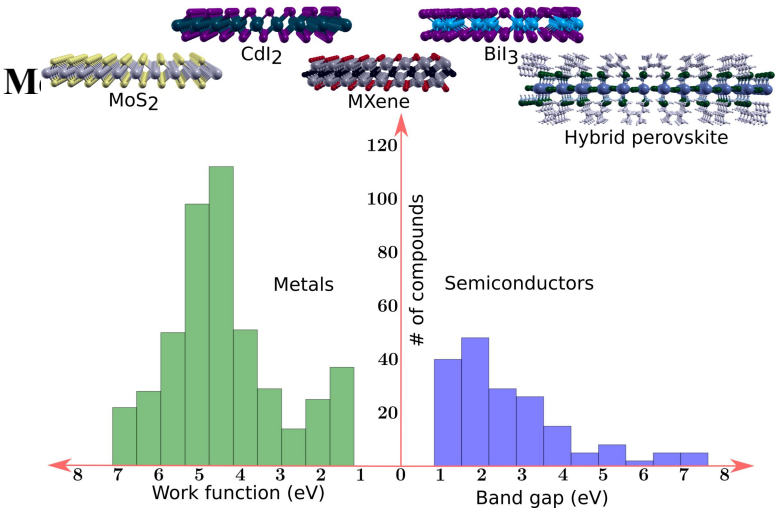


Figure 2. The 2D materials database maintained and developed in my group presently contains structural and electronic properties (band structures, dielectric functions) for around 500 2D materials of various types. The graph shows the distribution of the semiconductor/insulator band gaps and metal work functions. This and other materials databases will form an ideal starting point for the design of novel 2D materials and heterostructures with tailored light-matter interactions.

The project will apply first-principles electronic structure calculations at the level of density functional theory

(DFT) and many-body perturbation theory (GW and the Bethe-Salpeter Equation).

The calculations will mainly be performed using the well equipped and highly flexible GPAW[42] open source electronic structure code and the Atomic Simulation Environment (ASE)[46]. The ASE is a Python-based scripting environment with interface to about 20 different electronic structure codes including VASP, Quantum Espresso, FHI-Aims, Siesta, ABINIT, DFTB+, etc.. The methods developed in the course of the project will be implemented in GPAW and thus be accessible to the entire community. Via the tight coupling to the ASE it will be possible to use large parts of the developed code in combination with the other codes supported by ASE.

Defect formation enthalpies and charge transition energies will be calculated using plane waves or real space grids and the standard method of charged supercells. The PBE functional will be employed as a standard for total energies and structure optimizations while HSE calculations will be performed for validation and when higher accuracy is required. Initial and final state potential energy surfaces for transitions to defect levels will be calculated using constrained DFT as implemented in the GPAW code[68]. Band structures of pristine crystals will be calculated by the G_0W_0 method (when possible), the Heyd-Scuseria-Ernzerhof (HSE)[47] or the GLLBSC[48] xc-functionals for larger systems. It should be noted that GW band structure (and BSE optical spectra) calculations for 2D materials are non-trivial and require the use of a truncated Coulomb interaction, which in turn leads to an extremely slow k-point convergence due to the peculiar screening in 2D. I was the first to point out this issue[49]. Recently, I have developed a method that exploits an analytical form of the dielectric function to significantly speed-up GW calculations for 2D materials without compromising accuracy[50]. Phonons and electron-phonon couplings for defect supercells will be evaluated using GPAW's numerical atomic orbital basis set as I have previously done for calculations of phonon-limited mobilities in 2D materials[51].

Research plan

WP1: First-principles methods for non-radiative recombination- and carrier capture rates.

The recombination dynamics of carriers within the bulk of a semiconductor will be described using the semi-classical Boltzmann equation. In the absence of spatial and temporal gradients, the time evolution of the distribution functions (f_{nk} for Bloch states and f_i for localized defect states) is determined by electron-electron scattering (Auger and impact ionization) and electron-phonon scattering (Shockley-Read-Hall carrier capture). The various processes are illustrated in Figs. 3 and 4, respectively.

In steady state the scattering in and out of all microstates should cancel. Since the scattering rates depend on the distribution functions, this yields a set of non-linear equations. A simplified description may be obtained by assuming a quasi-equilibrium steady state defined by a quasi Fermi level for electrons and holes,

$$f_{c/v,k} = 1/(1 + \mathrm{e}^{(\varepsilon_{c/v,k} - \mu_{e/h})/kT})$$

where *c* and *v* denote conduction and valence bands, respectively. One then considers the change in the total population of excited electrons and holes, $n = \sum_c f_{ck}$ and $p = \sum_v f_{vk}$ rather than following the dynamics of the individual microstates. The occupation of localized defects states should be treated separately. It is customary to express the dynamics on a form that highlights the dependence of the various scattering processes on the carrier populations,

$$(1) \quad \frac{dn}{dt} = G - \sum_i A_i\, nc_i(1 - f_i) - Bnp - Cnnp - C'ppn - \sum_i D_i\, c_i nn(1 - f_i) - \sum_i D_i'\, c_i np(1 - f_i) + E$$

with a similar equation for the hole carriers. c_i and f_i denote the concentration and electron occupation number of defect *i*. *G* is a source term describing constant photogeneration of carriers. A_i are related to the one-body non-radiative carrier recombination, which proceeds through crystal defect levels as intermediate states – the so-called Shockley-Read-Hall (SRH) process, see Fig. 3. The two-body coefficient *B* describes radiative recombination of free carriers and is essentially given by the dipole matrix element between the initial and final states. *C* and *C'* are the three-body band-to-band Auger recombination coefficients, see Fig. 4(a,c). D_i and D_i' represent defect-assisted Auger carrier capture, see Fig. 4(b), and *E* describes carrier generation by impact ionization, which is independent of carrier concentration and essentially is an inverse Auger process.

Eq. (1) is merely a definition of the various recombination coefficients. To obtain these in practice requires calculation of transition matrix elements between all possible initial and final microstates. For example, the third term on the rhs (radiative recombination) must be calculated as

(2) $$Bnp = \sum_k \sum_{v,c} f_{ck}(1 - f_{vk})|\langle \psi_{ck}|\mathbf{r}|\psi_{vk}\rangle|^2 \, F(\varepsilon_{ck} - \varepsilon_{vk})$$

where F is the density of electromagnetic modes. The calculation of the remaining non-radiative rates is briefly described below.

Shockley-Read-Hall (SRH): The A_i coefficient corresponding to electron capture by a given defect in a given charge state, will be computed as follows. First, the phonon modes on the initial and final potential energy surfaces (PES) are evaluated using a supercell sufficiently large to avoid defect-defect interactions. Next, electron-phonon coupling matrix elements are obtained for phonons and electronic states corresponding to the initial state configuration R_i. The perturbation driving the transition is the electron-phonon (e-ph) coupling

$$H_{e-ph} = \sum_{k,nm} \sum_{\alpha,q} g^{\lambda}_{nk,mk+q} c^{+}_{kn} c_{k+q,m} \, Q_{\lambda,q}$$

where $g^{\lambda}_{nk,mk+q}$ are the e-ph coupling matrix elements and $Q_{\lambda,q}$ is a nuclear displacement operator. From Fermi's Golden rule it is then straightforward to obtain the energy conserving transition amplitudes from an initial product state consisting of an electron in the conduction band and a vibrational state on the initial PES to a final state consisting of a localized electron on the defect and an excited vibronic state on the final PES, see Fig. 3. Here, my previous experience with implementation of Boltzmann carrier transport with electron-phonon coupling will be useful[51].

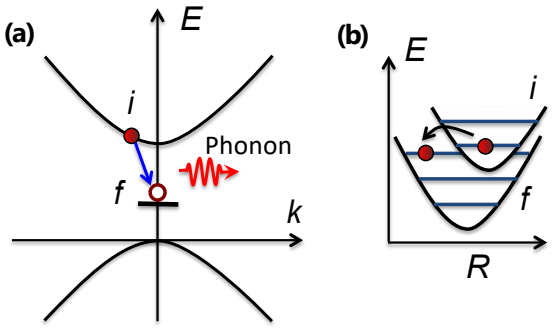


Figure 3. The Shockley-Read-Hall carrier capture process illustrated in a band diagram (a) and as transitions between different vibrational states on the initial and final state potential energy surfaces (b).

Auger scattering. The Auger scattering is driven by the screened Coulomb interaction and thus the relevant transition matrix elements entering the Fermi Golden rule are of the form $\langle \psi_1 \psi_2 | W | \psi_3 \psi_4 \rangle$. Care must be taken to account for spin and anti-symmetri of the many-body states but this is mainly a matter of book keeping. For defect-assisted Auger, one of the four electronic states is a localized level. The phonon-assisted Auger, is a second order process and thus the Coulomb matrix element is multiplied by an e-ph coupling ($g^{\lambda}_{nk,mk+q}$) matrix element. To simplify matters, W will be approximated by the statically screened Coulomb interaction evaluated in the random phase approximation (RPA). Moreover, as typical carrier concentrations will be low, most states involved in the Auger process will be close to the band extrema and it should be a good approximation to use the state *at* the band extrema for evaluating the matrix elements. This approximation will simplify the calculations considerably.

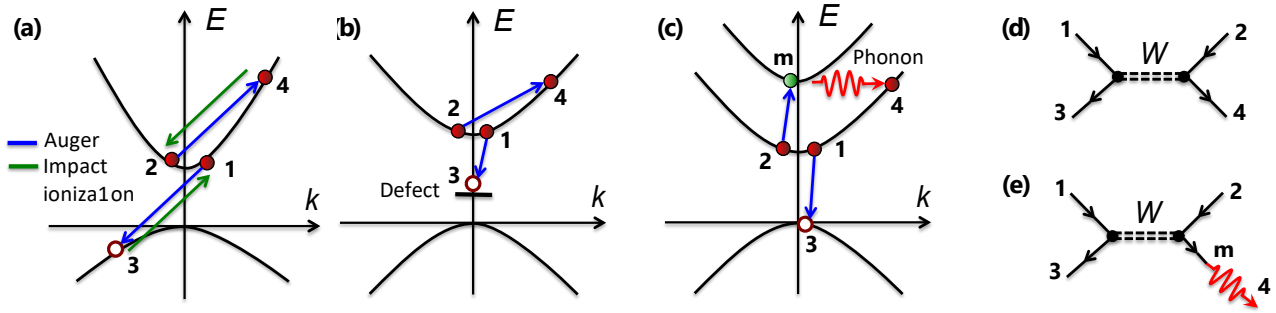


Figure 4. Different non-radiative processes in a semiconductor mediated by the (screened) electron-electron interaction. (a) Direct Auger recombination and impact ionization. (b) Defect-assisted Auger scattering. (c) Phonon-assisted Auger recombination. The second order process proceeds via the intermediate state m. (d) and (e) Examples of Feynman diagrams for the first-order Auger and Auger-phonon scattering processes.

Impact ionization. Referring to Fig. 4(a), impact ionization refers to a process where a highly excited electron in the conduction band (particle 4) decays to a lower energy state (2) and promotes an electron from the valence band (3) to the conduction band (1). The rate for this process can in fact given by the imaginary part of the GW self-energy of the pristine semiconductor in the ground state (at least for sufficiently low carrier concentrations) which can be obtained the existing implementation of the GW method in the GPAW code[52].

Computational procedure. For a complete characterization of the recombination dynamics for a given material, the first step is to perform a defect analysis. This involves calculating the formation energy and the charge transition energies for the most common, i.e. stable, defects. Based on the defect formation energies, the concentration of each defect, (c_i) is obtained, assuming an equilibrium Boltzmann distribution. Finally, the relevant matrix elements must be calculated as outlines above. With these inputs, the steady state carrier densities (n,p) under given irradiation conditions (G) can be calculated.

While all previous work has focused on one particular recombination process for one particular semiconductor/defect, I will apply the formalism systematically considering several different processes for many different materials (3D and 2D) and defects. This will allow me to establish a connection between the atomic and electronic structure of a material/defect and the carrier dynamics, which is essential to guide the design of materials to control light-matter interactions.

WP2: Predicting carrier lifetimes and PV performance of realistic photo-absorber materials

This work package will employ the methods developed in WP1 to perform a first-principles based evaluation of the *true* PV performance of realistic absorber materials by accounting for non-radiative recombination losses. This objective is well aligned with the experimental activities in the V-SUSTAIN center on the development of new absorber materials for photo-catalytic water splitting, and significant synergy effects are expected.

PV performance. One of the key performance parameters for PV materials is the open circuit voltage (V_{oc}). The V_{oc} is given by the difference in chemical potential of electrons and holes under solar irradiation, i.e. the difference between the electron and hole quasi Fermi levels. For an intrinsic semiconductor we have

$$V_{oc} = E_{F,e} - E_{F,h} = kT\ln(np/n_{eq}p_{eq})$$

where n (p) and n_{eq} (p_{eq}) are the electron (hole) carrier densities under illumination and in the dark, respectively. The non-equilibrium steady state carrier densities can be calculated using the methods developed in WP1. Since first-principles calculations of photovoltages have never been attempted before, we shall start by considering a set of simple and well characterized bulk materials. Specifically I shall consider Si and GaAs, which are known to produce large V_{oc} (approximately 0.2 eV below the band gap) and SnS, Cu_2O, which produce relatively low V_{oc} (around 0.6 eV below the band gap)[1].

The results will be analyzed with regard to which factors determine V_{oc}. In particular, which atomic/electronic structure features distinguish materials with low and high V_{oc}. Obviously the defect concentration is expected to play a major role. But perhaps more interestingly (in particular since the assumed equilibrium distribution of defects might not be realistic for many samples), what characterizes defects that are efficient in recombining charge carriers from defects that are not? How sensitive are the results with respect to the position of the defect level inside the gap? For SRH carrier trapping it is usually assumed that only deep gap states close to the band gap center are the most effective recombination centers. We shall put that hypothesis to test. How important are the structural relaxations between the occupied and empty defect level, i.e. the shift of the initial and final state potential energy surfaces? (see Fig. 10).

Defect tolerance. The term "defect tolerant" is often used loosely for semiconductors whose opto-electronic properties are not strongly influenced by the presence of defects[53]. I have recently shown that some of the transition metal dichalcogenides (TMDs) do not form deep gap states upon creation of the most common defect (sulphur vacancies[54]) indicating that these materials should be defect tolerant[55]. This project will provide a more elaborate analysis of the concept of defect tolerance beyond the simple "do common defects form deep gap states or not".

2D materials. The photophysics of the 2D semiconductors is distinctly different from the 3D case. In particular, quantum confinement and the suppressed and highly non-local dielectric screening in 2D materials leads to a rich set of excited states with unusual characteristics. I will explore how these properties manifest themselves in the non-radiative processes of these materials. My hypothesis is that defect-assisted processes will be enhanced in 2D materials because of the weak dielectric screening and strong wave function confinement.

There are several indications that this is the case, but a microscopic understanding is missing. For example, optical pump-probe experiments show that the carrier recombination dynamics in monolayer MoS_2 involve two very different time scales. On the basis of model calculations, it has been proposed that the two time scales originate from defect-assisted recombination via two different types of defects[56], see Fig. 5. In general, the monolayer TMDs exhibit rather poor quantum efficiencies below 0.01[57,58] (compared to >0.7 for the best III-V bulk semiconductors[59]) indicating that most electron-hole pairs recombine non-radiatively.

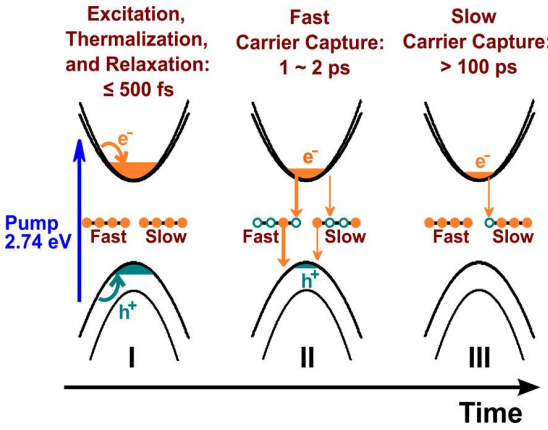


Figure 5. Optical pump-probe experiments for monolayer MoS_2 have shown that the transient dynamics of non-radiative carrier recombination has a fast and a slow time scale. On the basis of semi-empirical modelling, Auger recombination via two types of defects has been proposed as explanation. Figure from Ref. 56.

By performing a full characterization of the non-radiative recombination processes in monolayer MoS_2 (as described in WP1) I will elucidate the carrier dynamics in this prototypical 2D semiconductor. Comparing to bulk MoS_2 and the other bulk semiconductors, the differences between carrier dynamics in 2D and 3D materials will be elucidated. Note that bulk MoS_2 has an indirect band gap whereas monolayer MoS_2 has a direct gap. This difference is associated with quantum confinement and its effect should of course be distinguished from the effect of reduced dielectric screening.

WP3: Band structure engineering in van der Waals heterostructures

Multi-junction solar cells combine different photoabsorbers with carefully matching band gaps and band alignments to maximize solar light absorption and minimize thermalization losses, see Fig. 6. The vdWHs represent a novel materials platform which could potentially lead to the realization of atomic-scale band structure engineered devices with high flexibility at low cost. However, there are presently no methods for predicting and guiding the design of vdWHs and experimentalists are left to trial and error. Given the huge size of the vdWH compound space and time consuming process of synthesizing them, it is absolutely essential for the continued success of vdWH science, to develop efficient and reliable methods for calculating their electronic structure.

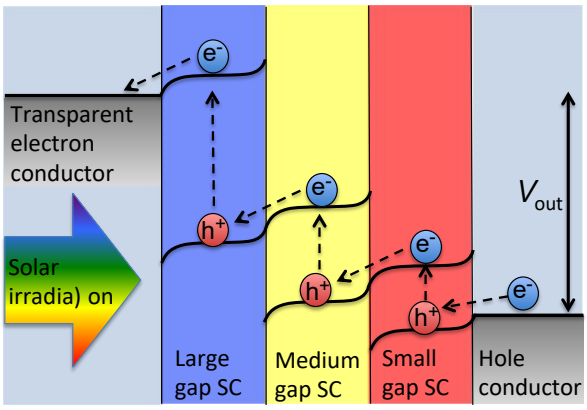


Figure 6. A multi-junction solar cell combines different semiconductors (SC) with band gaps and band alignments carefully optimized for maximal solar light absorption and minimal thermalization losses. For simplicity the quasi-Fermi levels are not shown although they define the required band alignment.

Band structure calculations for vdWHs is an outstanding challenge, not least due to the incommensurate nature of the heterointerfaces, which render conventional supercell calculations impossible, see Fig. 7(a). Accurate band structure calculations must go beyond standard mean field theories and solve the quasiparticle equation with a self-energy accounting for many-body effects. The many-body GW approximation is the state of the art, and has been shown to yield accurate band gaps for a variety of materials, including 2D materials[22,23]. However, the GW method is computationally very expensive and limited to materials with up to 20 atoms in the unit cell. Conventional GW calculations are thus completely unfeasible for vdWHs that can easily contain thousands of atoms in a unit cell.

The QEH and GΔW methods. I have recently developed a hybrid quantum/classical methodology that takes the first step to overcome this problem: The quantum electrostatic heterostructure model[24] (QEH) computes the dielectric function of a general vdWH from the ab-initio calculated dielectric functions of the individual layers assuming purely electrostatic coupling between the layers, i.e. neglecting interlayer hybridization. The accuracy of the QEH has been thoroughly documented[24,60]. The QEH method can be used to calculate the change in the screened Coulomb interaction (ΔW) of a given layer in the stack due to the surrounding layers. This in turn allows us to obtain the change in the GW electron self-energy, and thus the

change in the band energy relative to the freestanding layer. This non-local screening effect can by significant (on the order of 1 eV) but is completely missed by all known xc-functionals. The resulting GΔW method thus incorporates the effect of interlayer screening (correlation) on the band energies of a 2D material embedded in a vdWH without having to represent the different layers in the same supercell. Importantly, I have also demonstrated that the effect of interlayer hybridization can be treated separately and at a lower (single-particle) level of theory such as DFT.

The QEH and GΔW techniques have been used to compute the band structure of multilayer MoS_2 films with 1-100 layers yielding excellent agreement (within 0.1 eV) with the first-principles GW band structure of bulk MoS_2 for the thickest slabs (unpublished). In another study, I calculated band alignment and excitonic transition energies for a MoS_2-WSe_2 heterobilayer with a varying number of hBN spacer layers between. The results of the GΔW method are in excellent agreement (<0.15 eV) with the experimental photoluminescence (PL) spectra for both interlayer and interlayer excitons, see Fig. 7(c). In comparison the band structure is obtained from DFT with the LDA or the HSE xc-functionals underestimates the exciton energies by around 0.5 and 1.0 eV, respectively. These are only the first encouraging results and much more awaits to be explored, but requires a further extension of the methodology as described below.

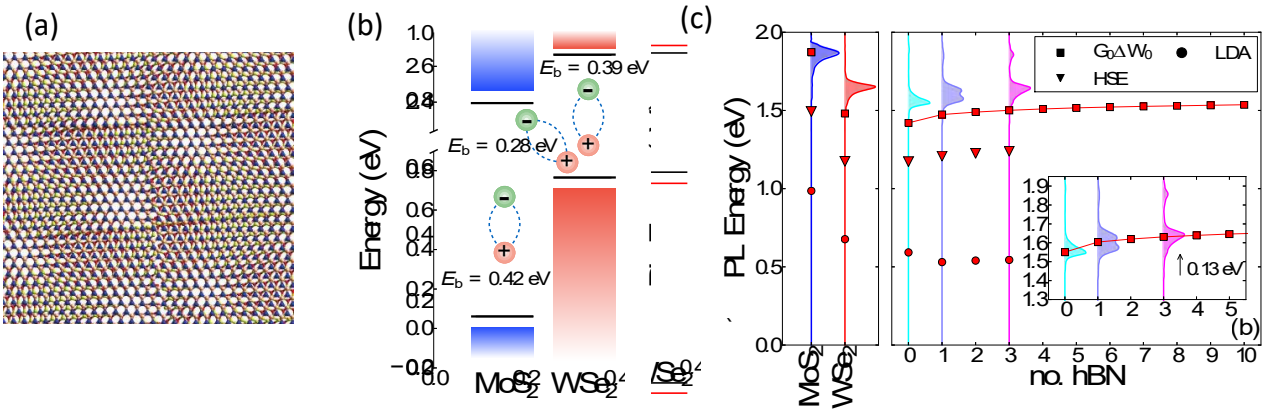


Figure 7. (a) Heterobilayer forming a complex superstructure (moire pattern) due to lattice mismatch and rotational offset of the two layers. (b) Band alignment and binding energies of the lowest excitons at a MoS_2/WSe_2 heterobilayer calculated using the G Δ W and QEH methods. (c) The calculated photoluminescence peak (obtained as the band gap minus the exciton binding energy) is plotted as a function of the number of intercalated hBN spacer layers, and compared with experiments (filled curves). Excellent agreement with experiments is obtained when the band structure is calculated using the GΔW method (squares). Figure from Ref. 25.

Beyond the GΔW method. What is missing in the GΔW methodology is the inclusion of interlayer hybridization and charge redistribution effects such as Pauli repulsion. These effects can shift and bend the bands, the latter being important to achieve carrier separation with the absorber layers. In the two examples mentioned above, hybridization and charge redistribution was either simple to include (multilayer MoS_2) or was judged to be insignificant (MoS_2/hBN/WSe_2).

To account for all three effects (interlayer screening, interlayer hybridization, and charge redistribution) I will combine the GΔW method with DFT, or tight-binding (DFTB) for very large systems as implemented in the dftb+ code[61]. For DFT with semi-local xc-functionals it is realistic to treat systems with up to 1000 atoms per unit cell. DFTB with standard parameters calculate the single-particle band structure self-consistently for a cell with one thousand atoms in a few minutes on a single CPU. By manipulating the basis functions underlying the DFTB model or the pseudopotentials used in DFT, the band structures of the *isolated* 2D materials of the stack can be made to agree with the corresponding GW band structures. A similar strategy has recently been successfully to obtain accurate band structures of complex Si-MOSFETs[62]. The band structure of the entire vdWH can then be computed self-consistently at the DFT/DFTB level, adding the GΔW self-energy corrections to account for the many-body interlayer screening. The DFTB-GΔW method will be developed and implemented in collaboration with Prof. Thomas Niehaus (University of Lyon) who is an expert developer of the dftb+ code.

The DFT(B)-GΔW method will allow us to compute band structures of general vdWHs containing hundreds of 2D layers and with complex lateral unit cells to account for misalignments and lattice mismatch. The calculations will require the GW band structure and dielectric functions of the freestanding versions of all the

2D layers in the stack. This input data can be calculated and stored once and for all. I have already constructed a database with such data for around 50 monolayer TMDs[40]. In the present project, these efforts will be extended significantly starting with the 300 simplest (<6 atoms per unit cell) experimentally known structures identified by Mounet *et al.*[7]. The resulting database will in itself represent a most useful outcome of the project.

Computational design. As a first application, the method will be used in conjunction with a genetic algorithm to explore the fundamental limits for vdWH band structure engineering. Is it really possible to design the band structure profile of a vdWH with atomic-scale precision, i.e. layer by layer, and what are the restrictions (if any)?

Next, I will consider the design of multi-junction solar cells. To harvest sufficient sunlight each absorber must contain at least 5 monolayers (a single MoS_2 absorbs around 10% of the incoming solar spectrum above the band gap[63]). This implies that the absorption spectrum of each absorber, to a first approximation, can be assumed to equal that of the bulk crystal, which can be readily computed, e.g. using the many-body Bethe-Salpeter equation. For tandem, triple and quadruple junctions I will consider absorbers of 5,15, and 30 monolayer thickness composed of 2D materials picked from a pool of 20-50 semiconductors. The theoretical cell efficiency for all possible absorber combinations will be evaluated (assuming 100% quantum efficiency) using high-throughput DFT(B)-GΔW band structure calculations.

WP4: Computational discovery of "isolated band metals" for low-loss plasmonics

An isolated band metal (IBM) is a material with one or several partially occupied bands separated from all other bands by finite energy gaps, see Fig. 8. An IBM with optimally designed band widths and band gaps, would be a "dream material" for plasmonics. For this, the plasma frequency and the band gaps should be as large as possible for the IBM to be metallic (negative permitivity) in the frequency range where the absorption is zero. Under these conditions, plasmon-polaritons with extremely large lifetimes could in principle be achieved. IBMs would also be ideal for transparent conductors. In this case, the optimal IBM should have a high density of free electrons in the isolated band(s) to facilitate a high conductivity but still low enough that the plasma frequency does not become too large as it defines the frequency onset of transparency. In addition, the band gaps should be as large as possible for a high onset of absorption and thus maximal window of transparency.

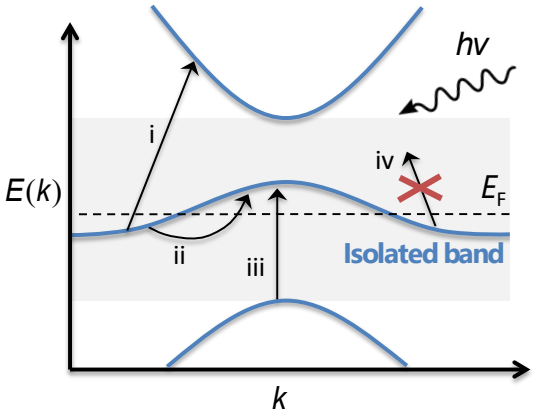


Figure 8. Optical losses in metals originate from electronic transitions in the crystal band structure (i, ii, iii). A metal with an isolated band will have reduced number of transitions available at specific frequencies (iv). When the gaps are sufficiently large to separate intraband (ii) from interband (i, iii) transitions, the optical losses will be strongly reduced in the corresponding frequency range.

To the best of my knowledge, the discovery of an IBM has never been reported. There is, however, no fundamental reason why such metals should not exist. Low-dimensional materials, in particular layered or monolayer materials, with a significant fraction of under-coordinated atoms and thus overall weaker hybridization, might be more likely to exhibit such peculiar band structures.

Based on this hypothesis I recently discovered that the transition metal dichalcogenides (TMDs) 2H-NbS_2 and 3R-TaS_2 are nearly IBMs in their layered bulk form and true IBMs in their monolayer form. Note, that the monolayer forms of these materials have recently been isolated and their (normal) metallic nature at room temperature confirmed[64]. ARPES measurements on monolayer TaS_2 clearly shows the isolated nature of the occupied part of the band structure[65]. Our calculated figures of merit (plasmon lifetimes, propagation length, and spatial confinement) show that these metals are comparable to, or even outperform the best known plasmonics material silver, at least in thin film geometries[66]. This is a direct consequence of the suppressed losses due to the special band structure. To reduce losses further (for plasmonics) and improve the transmittance (for transparent conductors) the band gaps around the isolated band must be increased.

Discovery of novel IBMs. To discover new and superior IBMs, we will screen the band structures of the 1800 experimentally known layered bulk materials[7]. It is stressed that while the crystal structure of these 1800 materials is known, their optical properties have, for the majority of materials, never been investigated. The screening will be performed using DFT-PBE calculations, which are efficient and qualitatively reliable.

An initial screening of the 200 simplest materials (<6 atoms per unit cell) already identified ten IBMs including TaS_2 and NbS_2. The electronic and optical properties of these candidate IBMs will be investigated in depth using DFT-HSE and GW (when possible). To obtain a realistic description of the optical permittivity it is imperative to account for scattering (by defects, impurities, phonons, electrons). Such scattering processes lead to losses via single-particle transitions that are not allowed in the pristine crystal due to momentum conservation, processes (i) and (ii) in Fig. 8. In my previous work[66] I used the ab-initio calculated RPA response functions with an energy-dependent relaxation rate of the form $\gamma(\varepsilon) = a\mathrm{JDOS}(\varepsilon)$, where a is a parameter controlling the scattering strength and JDOS is the density of states for electron-hole pairs (joint density of states). Compared to the standard constant relaxation time approximation, this model accounts for band structure effects in the damping term, which is essential for materials with non-trivial band structures like the IBMs. The parameter a was chosen such as to reproduce the experimental line width of the Ag(111) surface plasmon-polariton. Using the methods developed in WP1 (generalized to metals) it will be possible to go beyond the JDOS model and put it to test, by explicitly accounting for electron-phonon or defect scattering.

In parallel with the in-depth investigations of the experimentally known candidate IBMs, completely new materials will be explored using chemically guided, combinatorial substitution of the elements for each candidate IBM. In my previous work[66] I successfully applied this strategy to NbS_2 and TaS_2. Specifically, I explored MXY structures obtained by replacing the group 4 metals (V,Nb,Ta) by group 3 metals (Ti,Zr,Hf) and every second S by a halide (Cl,Br,I). These replacements conserve the overall valency, crucial for the chemical stability of the material, but were found to increase the energy gaps due to the lower electronegativity of the group 3 compared to group 4 metals and higher electronegativity of the halides compared to chacolgenide S. As a consequence, the plasmonic figures of merit were further improved.

The application of this design strategy to the entire set of candidates resulting from the screening of experimentally known compounds, is likely to disclose even better IBMs. If successful, these developments could have game changing consequences for the field of plasmonics and photonic materials, which are presently stuck due to the high intrinsic losses of the known metals. As basis material for transparent electrodes, the IBMs could become an interesting crystalline-metal alternative to the heavily doped oxides, with potentially lower cost and the possibility of hole conduction.

WP5: High-throughput computational discovery of novel 2D color centers.

The recent discoveries of single photon emitters (SPEs) in few-layer TMDs[37,38] and hBN[39] represent the first steps towards the realization of 2D SPEs. However, the observed SPEs are still far from ideal and suffer from low yield and poor stability, and the nature of the color centers (atomic and electronic structure) are not known. While 2D materials in principle offer an ideal platform for generating and manipulating quantum light, concrete realizations in terms of specific materials/defects are yet to be developed.

I will perform high-throughput DFT calculations to screen a large number of color centers in different 2D materials. Both intrinsic point defects as well as impurity atoms embedded in hBN, the group 4 TMDs, and other large band gap 2D materials like GaS and GaSe, will be considered. The potential of the color centers will be evaluated based of the four design criteria listed below. The criteria are listed in an order of increasing computational complexity. Color centers will only be subject to criterion *i* if they passed criterion *i*-1.

a. Stability: Structural stability is essential to avoid bleaching and frequency drift of the SPE. For SPEs embedded in a 2D material chemical stability becomes an issue if the defect site is exposed to the atmosphere. A high structural and chemical stability is correlated with a low defect formation energy and low binding energy of atoms such as H and O on the defect site, respectively. These quantities will be calculated using DFT. It should be noted that encapsulation of the 2D host material e.g. in hBN, could be a way to protect the defect from reacting with the environment. Such a procedure was successfully demonstrated for phosphorene[67]. If this option is considered, chemical stability becomes less critical.

b. Indistinguishability: Applications involving entanglement require indistinguishable photons. Coupling to the thermal bath of lattice vibrations and stray fields from the environment can lead to variation in photon energy, phase, and polarization thus lowering the degree of indistinguishability. To avoid variation in emission frequency due to stray fields, the charge density of the relevant electronic transition, $\rho(\boldsymbol{r}) = |\psi_i(\boldsymbol{r})|^2 - |\psi_f(\boldsymbol{r})|^2$, should have zero dipole moment. To minimize thermal fluctuations, the initial and final state potential energy surfaces should be well aligned, see Fig. 10. The total change in atomic coordinates, $\Delta R = \sum |\Delta R_n|$, between the initial and final states, provides a measure of this shift that can be calculated using constrained DFT.

c. Frequency: Ideally, the emission frequency should lie in the red/near infrared part of the spectrum, where excellent photo detectors and low loss optical waveguides are available, or in the optical telecommunication range (around 1.5 μm or 0.8 eV) for maximal compatibility with silica optical fibers. The emission frequency (corresponding to the zero phonon line) will be computed using the delta-SCF method[68] in GPAW.

d. Quantum efficiency: The quantum efficiency is the probability that an excited emitter produces a photon, i.e. $\Gamma_{rad}/(\Gamma_{rad}+\Gamma_{non-rad})$. A high quantum efficiency is important for reliable operation ("on-demand" photon sources). The radiative and non-radiative recombination rates will be calculated for the most promising color centers using the methodology developed in WP1.

WP6: 2D color center engineering

The single most important property of a solid state SPE is stability. Consequently, it is important to explore strategies for improving the remaining properties (b-d) of the most stable SPEs identified in WP5. This WP will focus on (i) the design of *novel hyperbolic materials* with tailored light dispersion and photonic density of states for enhancing the radiative recombination of the SPE and collecting the emitted photons, and (ii) the use of *strain engineering* for tuning the SPE frequency, spectral line width, and quantum yield.

Novel hyperbolic materials. A hyperbolic material[69] is characterized by having dielectric properties ($\varepsilon>0$) in one direction and metallic properties ($\varepsilon<0$) in two orthogonal directions (or vice versa). Traditionally, this has been realized by stacking alternating layers of metal films and dielectrics to form an artificial metamaterials with subwavelength lattice parameter in the range 20-100 nm. The extreme anisotropy leads to hyperbolic light dispersion with a range of possible applications including enhancement of spontaneous emission[70], negative refraction and hyperlensing[71].

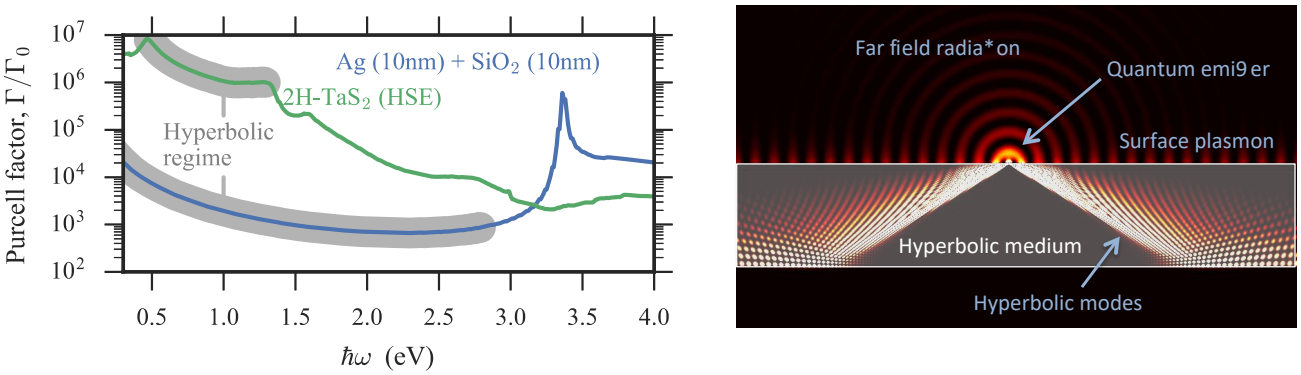


Figure 9. Left: Purcell factor (enhanced spontaneous emission rate relative to free space) of an emitter placed 1 nm above a conventional hyperbolic metamaterial (blue) and the natural hyperbolic material 2H-TaS_2 (green). The extremely large Purcell factor of 2H-TaS_2 is a result of its homogeneous nature (as opposed to the artificially structured metamaterials), which reduces scattering losses and increases the number of hyperbolic modes dramatically. Right: Simulated field energy distribution around a radiating point dipole (quantum emitter) positioned 1 nm above the surface of a thin hyperbolic slab. Most of the radiated field (>90%) is directed into the hyperbolic film from which it can escape only at the ends. Unpublished results from my group.

I have recently discovered that most of the layered TMDs are *naturally hyperbolic* as a result of their layered structure which leads to an intrinsic anisotropy. Remarkably, because of their homogeneous nature, these natural hyperbolic materials have much higher photonic densities of states than the conventional artificially structured metamaterials, see Fig. 9(left). Recently, hBN was shown to be hyperbolic in its Reststrahlen band between the longitudinal and transverse optical phonon branches[72] and graphite has been reported to be hyperbolic in the UV[73]. My findings indicate that hyperbolic dispersion is a general property of van der Waals layered crystals. Most interestingly, I find that *metallic* 2D layered materials can be hyperbolic in the range between the in-plane and out-of-plane plasma frequencies. For example, in the case if 2H-TaS_2 the hyperbolic regime extends from 0.4 eV – 1.4 eV. This opens unique possibilities for controlling SPEs and could ultimately form the basis for an electrically controlled SPE device, see WP7.

This WP will pursue the design of vdWHs with tailored light dispersion and density of states. In particular, heterostructures and/or pristine layered crystals with peaks in the photonic density of states at frequencies matching the most promising 2D SPEs discovered in WP5 will be identified.

Placing a 2D SPE ontop of a hyperbolic substrate, not only permits a many-fold enhancement of the radiative emission rate, but also provides an efficient means of collecting the emitted photon. Indeed, it has been shown that the fraction of photons emitted into a hyperbolic substrate can exceed 90%, i.e. the photon emission becomes highly directional[74]. Furthermore, due to the special properties of the hyperbolic modes, the photon can escape the hyperbolic medium only through a surface perpendicular to the layers. For a thin film made of a hyperbolic layered material this means that the light can only escape through the periphery of the film, see Fig. 9(right). Optimizing the propagation length of the hyperbolic modes then becomes critical for extracting the emitted photons with high yield. To this end the 2D isolated band metals (IBMs) discovered in WP4 will be of particular interest because their special band structure entails lowered Ohmic losses and thus longer photon propagation lengths.

Strain engineering. Disregarding the effect of stray fields from the environment, which can lead to dephasing and frequency variation, the properties of a SPE are essentially determined by the shape and relative position of the potential energy surfaces (PES) of the initial and final electronic states, see Fig. 10 for a simplified 1D illustration. According to the Franck-Condon principle, the emitted photon energies correspond to the differences between the vibrational levels on the two PESs and the transition amplitudes involve an overlap of the initial and final vibronic wave functions. The ideal situation thus occurs when the PESs are well aligned so that transitions between different vibrational states are supressed (due to orthogonality) and most photons are emitted to the zero-phonon band. I will use constrained DFT calculations to explore the possibility of controlling the relative position of the PESs by applying strain to the 2D host material. My experimental collaborator, Kirill Bolotin, has previously demonstrated detailed control over the band structure and optical transitions in monolayer MoS_2 via tensile strain[75]. He will perform similar experiments for SPEs as part of his 2015 ERC-stg "Strained2DMaterials" (see attached letter of support). A successful outcome will require a close feedback loop between theory and experiments. To ensure this, resources have been allocated for several short term visits to Kirill Bolotin's group.

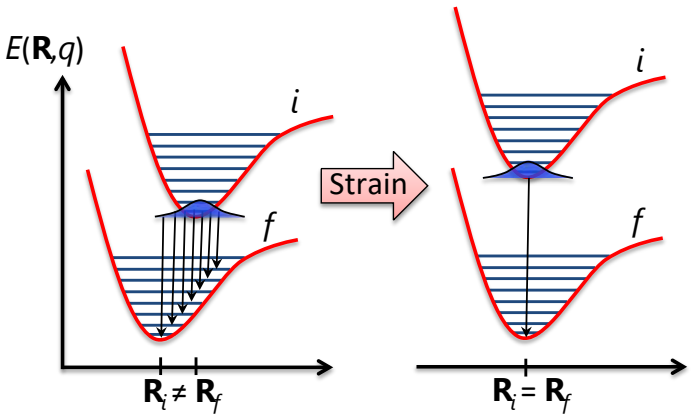


Figure 10. Potential energy surfaces corresponding to the initial and final electronic configurations of a SPE. The vibrational levels are indicated by horizontal lines. By straining the 2D host material, the relative position of the two surfaces can be shifted vertically and horizontally, which enables control over the emitted photon energy and linewidth, respectively.

WP7: Experimental characterization and realization of an electrically controlled SPE

Compact and integrated "on demand" single-photon light sources should be electrically controlled. The 2D SPEs are ideally suited for this purpose, as electrons can tunnel directly from the source electrode to the localized defect state, emit a photon, and tunnel onto the drain electrode, see Fig. 11. By choosing the electrodes as a hyperbolic layered metal one would obtain the added benefits of Purcell enhanced radiative emission and photon extraction via the hyperbolic modes. This WP will undertake the ambitious challenge of realizing such a dream device.

My experimental collaborator and colleague Prof. Ulrik L. Andersen (DTU-Physics) is leading a world class quantum optics lab specializing on SPE and quantum communication. He has extensive experience with NV centers and colloidal quantum dot SPEs by applying those as optical probes in nano-photonic and plasmonic systems. LIMA will fund a post doc who will work in Prof. Andersen's lab on the characterization of the proposed 2D SPEs, the natural hyperbolic metals, and ultimately the realization of an electrically controlled hyperbolic SPE device. My colleague Prof. Peter Bøggild (DTU-Nanotech) is an expert on 2D materials electronics and has developed an efficient procedure to produce ultraclean vdWHs from mechanically exfoliated 2D layers[76]. He will provide the 2D materials and vdWHs required for these activities.

As a first step, spectroscopic ellipsometry will be used to characterize the complex dielectric functions of the layered hyperbolic metals predicted by computations (e.g. 2H-TaS_2, 3R-NbS_2 and beyond, depending on the outcome of the computational discoveries). In a next step, we will use the hyperbolic material(s) as substrate to modify the radiative properties of well-characterized quantum emitters, such as colloidal core/shell InAs/InP quantum dots pre-selected for emission wavelength matching the hyperbolic regime of the substrate. By applying a spacer layer of known thickness and refractive index, the dependence of the emission rate and directionality[77] on the SPE-hyperbolic substrate separation will be explored using time resolved spectroscopy and imaging of the back focal plane (k-space) on a CCD camera.

The ultimate goal of this WP is the realization of an electrically controlled SPE as illustrated in Fig. 11. Selection of materials (2D SPE and hyperbolic electrode) will be made based on the outcome of the combined computational and experimental investigations in WPs 4-6. LIMA provides the ideal platform for undertaking this challenging and potentially ground breaking endeavour by providing a tight coupling between scientists with complementary skills in computational modelling, quantum optics, and 2D materials electronics.

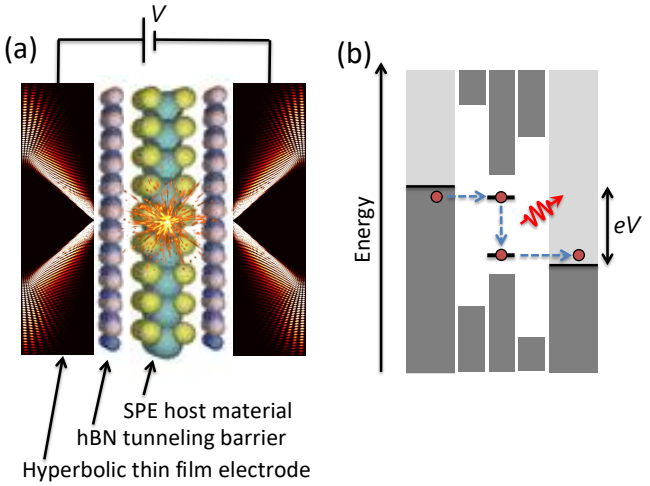


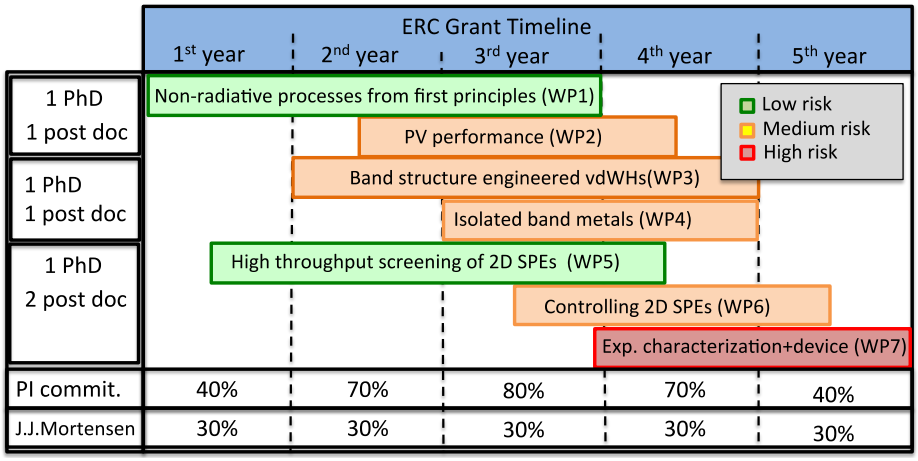
Figure 11. (a) The envisioned electro-hyperbolic SPE dream device. The 2D material hosting the color center is sandwiched between two metallic electrodes with hyperbolic light dispersion for enhanced emission rates and efficient photon collection. Thin tunnelling barriers, e.g. hBN layers, separate the SPE from the electrodes. (b) Energy level diagram of the device.

Impact

The proposed methods for calculating non-radiative recombination rates from first principles will facilitate a realistic theoretical assessment of the potential of candidate materials for photovoltaics, photocatalysis and light emitting diodes and thus open the door to computational design of new materials within these fields. The hybrid DFT-GW method for band structure calculations of realistic vdWHs will make it possible to scan the huge vdWH compound space and allow for 'in-silico' exploration of different heterostructure designs before undertaking the time demanding task of synthesizing the material. The isolated band metals are a completely new type of materials with the potential to reduce Ohmic losses in the NIR frequency regime far beyond the level of any presently known metal. The discovery of such a metal could have game changing impact on the field of plasmonics, which is extremely challenged by optical losses in the metallic components. Similarly, the discovery of a vdW layered IBM with low optical losses and hyperbolic dispersion in the same frequency range, combined with the possibility of tuning the light dispersion by stacking different 2D materials, would open completely new perspectives for photonic metamaterials. Finally, the discovery and demonstration of a useful 2D single-photon emitter would be important for a number of quantum technology applications. The proposed electrically controlled SPE with hyperbolic electrodes (electro-hyperbolic SPE) is a dream device made possible by the concepts and materials developed by LIMA. Its successful realization would represent a quantum leap for "on-demand" SPE technology.

Team and infrastructure

The core scientific team is comprised by myself, the hired 3 PhD students and 4 post docs, and GPAW code developer Jens Jørgen Mortensen from DTU-Physics (30% on the project). The project involves collaborations with Prof. Thomas Niehaus (University of Lyon) on the DFTB-GW method and three experimentalists with complementary world leading expertise on single-photon emitters



(Prof. Ulrik Andersen, DTU-Physics), 2D materials electronics (Prof. Peter Bøggild, DTU-Nanotech), and 2D materials strain engineering (Kirill Bolotin, Freie University Berlin).

The project will be carried out at the Department of Physics at the Technical University of Denmark. DTU-Physics manages the Niflheim supercomputer cluster[78] with >7000 CPU cores and a peak performance of >70 teraflops. To meet the substantial demands for computer power, it is part of this proposal that Niflheim is expanded by 26 compute nodes which will be dedicated to this project.

Section c. Resources (including project costs)

Cost Category			**Total in Euro**
Direct Costs	**Personnel**	PI	0
		Senior Staff	0
		Postdocs (x4)	578,559
		Students (x3)	492,550
		Other (Technical Code Developer, PhD eval. fee)	174,920
	i. Total Direct Costs for Personnel (in Euro)		1,246,029
	Travel		67,204
	Equipment		184,409
	Other goods and services	Consumables	38,441
		Publications (including Open Access fees), etc.	25,000
		Other (please specify)	
	ii. Total Other Direct Costs (in Euro)		315,054
A – Total Direct Costs (i + ii) (in Euro)			1,561,083
B – Indirect Costs (overheads) 25% of Direct Costs (in Euro)			390,271
C1 – Subcontracting Costs (no overheads) (in Euro)			0
C2 – Other Direct Costs with no overheads (in Euro)			0
Total Estimated Eligible Costs (A + B + C) (in Euro)			1,951,354
Total Requested EU Contribution (in Euro)			1,951,354

Please indicate the duration of the project in months:	60
Please indicate the % of working time the PI dedicates to the project over the period of the grant	60%

Personnel: Salary for 3 PhDs (total 9 man-years) and 4 post docs (total 8 man-years). The budget also covers 30% of the salary for Jens J. Mortensen whose skills in high performance code development is of key importance to the project.

Travels: Travels to two conferences/year for all scientific staff of the project. (20 man years, 1500 Euro per conference; in total 60 k€). The remaining 7k€ covers visits to/by Kirill Bolotin.

Publications: I expect that the project will result in around 20 scientific papers in international peer-reviewed journals providing open access (total costs 25 k€; 1600€ per paper).

Equipment: 26 dedicated compute nodes each containing 20 Intel ”Haswell" CPU cores, 128 GB of RAM memory, and Infiniband interconnect (150 k€). Single photon detector for NIR wave lengths (30 k€).

Open access: All generated data will be made accessible via the open Computational Materials Repository[43] and/or the open access NOMAD repository[45]. All developed methods will be implemented in the open source code GPAW[41] and made available to the public.

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